

CSS Easy Notes

Slide-note style guide with simple explanations and practical code examples

CSS means Cascading Style Sheets. HTML creates the structure of a web page. CSS designs that structure and makes it beautiful, readable, responsive, and professional.

- Use CSS to change colors, fonts, spacing, borders, layouts, buttons, cards, forms, animations, and mobile responsiveness.
- These notes are written so a beginner can read them and understand how CSS is used in real web pages.

1. What is CSS?

CSS is a styling language used to design HTML elements. For example, HTML can create a paragraph, but CSS can change its color, size, font, spacing, and position.

```
<p>This is a paragraph.</p>

p {
  color: blue;
  font-size: 20px;
}
```

In this example, the paragraph text becomes blue and its size becomes 20 pixels.

2. Basic CSS Syntax

CSS is written using a selector, property, and value. The selector chooses the HTML element. The property tells what you want to change. The value tells how you want to change it.

```
selector {
  property: value;
}

h1 {
  color: red;
  font-size: 40px;
}
```

This means all h1 headings will become red and their font size will become 40px.

3. Types of CSS

3.1 Inline CSS

Inline CSS is written directly inside an HTML tag using the style attribute. It is useful for quick testing, but it is not good for large projects because it makes HTML messy.

```
<h1 style="color: red;">Hello World</h1>
```

3.2 Internal CSS

Internal CSS is written inside a style tag in the head section of the HTML file. It is useful when you want to style only one page.

```
<head>
  <style>
    h1 {
      color: blue;
      text-align: center;
    }
  </style>
</head>
```

3.3 External CSS

External CSS is written in a separate .css file. This is the best method for real websites because HTML and CSS remain separate and easy to manage.

```
<!-- HTML file -->
<link rel="stylesheet" href="style.css">

/* style.css file */
h1 {
    color: green;
}
```

4. CSS Selectors

Selectors are used to select the HTML elements that you want to style.

4.1 Element Selector

```
p {
    color: blue;
}
```

This applies blue color to all paragraph tags on the page.

4.2 Class Selector

```
<p class="intro">This is introduction text.</p>

.intro {
    color: green;
    font-size: 20px;
}
```

A class starts with a dot in CSS. The same class can be used on many elements.

4.3 ID Selector

```
<h1 id="main-heading">Welcome</h1>

#main-heading {
    color: red;
    text-align: center;
}
```

An ID starts with # in CSS. ID should usually be unique on one page.

4.4 Universal and Group Selectors

```
* {
    margin: 0;
    padding: 0;
}

h1, h2, p {
    color: black;
}
```

The universal selector selects everything. The group selector applies the same style to multiple elements.

5. Colors in CSS

CSS can add colors to text, backgrounds, borders, buttons, and many other elements.

```
h1 {
  color: blue;
}

body {
  background-color: lightgray;
}

button {
  background-color: rgb(0, 120, 255);
}
```

CSS supports color names like red, HEX values like #333333, and RGB values like rgb(0, 120, 255).

6. Text Styling

CSS can control how text looks on a web page. You can change font size, font family, boldness, alignment, and line spacing.

```
p {
  color: #333;
  font-size: 18px;
  font-family: Arial, sans-serif;
  font-weight: bold;
  text-align: center;
  line-height: 1.6;
}
```

- font-size changes text size.
- font-family changes the font style.
- font-weight makes text bold or normal.
- text-align moves text left, right, center, or justify.
- line-height controls space between lines.

7. Font Properties

Font Size

```
h1 {
  font-size: 40px;
}

p {
  font-size: 18px;
}
```

Headings normally use bigger font sizes. Paragraphs normally use medium font sizes.

Font Family

```
body {
  font-family: Arial, sans-serif;
}
```

This tells the browser to use Arial. If Arial is not available, it will use another sans-serif font.

Font Weight

```
h1 {  
  font-weight: 700;  
}  
  
p {  
  font-weight: normal;  
}
```

700 is commonly used for bold text. normal is used for regular text.

8. CSS Box Model

Every HTML element is treated like a box. The CSS box model has four main parts: content, padding, border, and margin.

```
.box {  
  width: 300px;  
  padding: 20px;  
  border: 2px solid black;  
  margin: 30px;  
}
```

- Content is the actual text or image.
- Padding is the space inside the box.
- Border is the line around the box.
- Margin is the space outside the box.

9. Width, Height, Padding, Margin, Border

```
.card {  
  width: 300px;  
  height: 200px;  
  padding: 20px;  
  margin: 30px;  
  border: 2px solid black;  
}
```

Width and height control size. Padding adds inner space. Margin adds outer space. Border adds a line around the element.

Border Radius

```
.card {  
  border-radius: 15px;  
}
```

Border radius makes corners rounded. It is commonly used for cards, buttons, images, and input fields.

10. Background

```
body {  
  background-color: #f5f5f5;  
}  
  
.hero {  
  background-image: url("background.jpg");  
  background-size: cover;  
  background-position: center;  
}
```

background-color adds a plain color. background-image adds an image. background-size: cover makes the image cover the full area. background-position: center keeps it centered.

11. Display Property

The display property controls how an element appears and behaves on the page.

```
display: block;  
display: inline;  
display: inline-block;  
display: flex;  
display: grid;  
display: none;
```

- block elements take full width and start on a new line, like div and p.
- inline elements take only needed width, like span and a.
- inline-block allows width and height while staying inline.
- flex and grid are used for modern layouts.
- display: none hides the element.

12. Flexbox

Flexbox is used to align items easily in rows or columns. It is very useful for navbars, cards, buttons, and layouts.

```
.container {  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  gap: 20px;  
}
```

- display: flex activates flexbox.
- justify-content controls horizontal alignment.
- align-items controls vertical alignment.
- gap adds space between items.

Flex Direction

```
.container {  
  display: flex;  
  flex-direction: row;  
}  
  
/* row = side by side, column = top to bottom */
```

13. CSS Grid

CSS Grid is used for two-dimensional layouts. It works with rows and columns. It is useful for gallery sections, dashboards, and product cards.

```
.grid-container {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 20px;  
}
```

This creates three equal columns. 1fr means one equal part of the available space.

14. Position Property

Position controls where an element appears on the page.

```
position: static;  
position: relative;  
position: absolute;  
position: fixed;  
position: sticky;
```

Relative Position

```
.box {  
  position: relative;  
  top: 20px;  
  left: 30px;  
}
```

The element moves from its normal position.

Absolute Position

```
.parent {  
  position: relative;  
}  
  
.child {  
  position: absolute;  
  top: 10px;  
  right: 10px;  
}
```

The child is positioned according to its nearest positioned parent.

Fixed Position

```
.navbar {  
  position: fixed;  
  top: 0;  
  width: 100%;  
}
```

The element stays fixed on the screen even when the user scrolls.

15. Buttons in CSS

```
<button class="btn">Click Me</button>  
  
.btn {  
  background-color: #007bff;  
  color: white;  
  padding: 12px 24px;  
  border: none;  
  border-radius: 8px;  
  font-size: 16px;  
  cursor: pointer;  
}
```

CSS can make a simple HTML button look modern. Padding increases size, border-radius rounds corners, and cursor: pointer shows a hand cursor.

Hover Effect

```
.btn:hover {  
  background-color: #0056b3;  
}
```

Hover effect works when the mouse goes over the button. It makes the website feel interactive.

16. Forms in CSS

CSS is used to make forms clean, readable, and professional.

```
<form class="form-box">  
  <label>Name</label>  
  <input type="text" placeholder="Enter your name">  
  <button>Submit</button>  
</form>  
  
.form-box {  
  width: 300px;  
  padding: 20px;  
  border: 1px solid #ddd;  
  border-radius: 12px;  
}  
  
.form-box input {  
  width: 100%;  
  padding: 10px;  
  margin-bottom: 15px;  
  border: 1px solid #ccc;  
  border-radius: 6px;  
}
```

Labels describe fields. Inputs become easier to use when they have proper width, padding, borders, and spacing.

17. Card Design

Cards are used in modern websites to show services, courses, products, profiles, and blog posts.

```
<div class="card">
  <h2>HTML Course</h2>
  <p>Learn HTML from basic to advanced level.</p>
  <button>Start Learning</button>
</div>

.card {
  width: 300px;
  padding: 20px;
  background-color: white;
  border-radius: 15px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.1);
}
```

This creates a professional card with white background, spacing, rounded corners, and soft shadow.

18. Shadow, Opacity, and Z-Index

Box Shadow

```
.box {
  box-shadow: 0 4px 10px rgba(0,0,0,0.2);
}
```

Shadow adds depth. It makes boxes look lifted from the page.

Opacity

```
.image {
  opacity: 0.7;
}
```

Opacity controls transparency. 1 means fully visible. 0 means invisible. 0.7 means 70% visible.

Z-Index

```
.box1 {
  position: absolute;
  z-index: 1;
}

.box2 {
  position: absolute;
  z-index: 2;
}
```

The element with higher z-index appears on top. Z-index works with positioned elements.

19. Responsive Design and Media Query

Responsive design means the website adjusts properly on desktop, tablet, and mobile screens.

```
.container {
  display: flex;
  gap: 20px;
}

@media (max-width: 768px) {
  .container {
    flex-direction: column;
  }
}
```

On large screens, items appear in a row. On screens smaller than 768px, items appear in a column. This makes the design mobile-friendly.

20. Transition, Transform, and Animation

Transition

```
.btn {
  background-color: blue;
  transition: 0.3s;
}

.btn:hover {
  background-color: red;
}
```

Transition makes changes smooth instead of instant.

Transform

```
.card:hover {
  transform: scale(1.05);
}
```

Transform can move, rotate, or scale an element. This example makes a card slightly bigger on hover.

Animation

```
.box {
  width: 100px;
  height: 100px;
  background-color: blue;
  animation: moveBox 2s infinite;
}

@keyframes moveBox {
  from {
    transform: translateX(0);
  }

  to {
    transform: translateX(200px);
  }
}
```

Animation creates movement. @keyframes defines the starting and ending points of the animation.

21. Complete CSS Example

This example creates a responsive card section with a header, three cards, hover effects, and mobile-friendly layout.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Practice Page</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f6f8;
      margin: 0;
      padding: 0;
    }

    header {
      background-color: #007bff;
      color: white;
      text-align: center;
      padding: 30px;
    }

    .container {
      display: flex;
      justify-content: center;
      gap: 20px;
      padding: 40px;
    }

    .card {
      width: 280px;
      background-color: white;
      padding: 20px;
      border-radius: 15px;
      box-shadow: 0 4px 15px rgba(0,0,0,0.1);
      transition: 0.3s;
    }

    .card:hover {
      transform: translateY(-8px);
    }

    .btn {
      background-color: #007bff;
      color: white;
      padding: 12px 20px;
      border: none;
      border-radius: 8px;
      cursor: pointer;
    }
  </style>
</head>
</html>
```

```

        @media (max-width: 768px) {
            .container {
                flex-direction: column;
                align-items: center;
            }
        }
    </style>
</head>
<body>
    <header>
        <h1>Learn CSS</h1>
        <p>CSS is used to style websites.</p>
    </header>

    <div class="container">
        <div class="card"><h2>HTML</h2><p>HTML creates structure.</p><button class="btn">Learn More</button>
        <div class="card"><h2>CSS</h2><p>CSS makes websites beautiful.</p><button class="btn">Learn More</button>
        <div class="card"><h2>JavaScript</h2><p>JavaScript adds interactivity.</p><button class="btn">Learn More</button>
    </div>
</body>
</html>

```

In this example, body controls the whole page, header creates the top section, container uses flexbox, card creates modern boxes, button styles the buttons, and media query makes it responsive on mobile.

22. Important CSS Properties Summary

Property	Use
color	Changes text color
background-color	Changes background color
font-size	Changes text size
font-family	Changes font style
font-weight	Makes text bold or normal
text-align	Aligns text
line-height	Adds space between text lines
width / height	Sets element size
padding	Adds space inside element
margin	Adds space outside element
border	Adds border around element
border-radius	Rounds corners
box-shadow	Adds shadow
display	Controls layout behavior
flex	Creates flexible row/column layout
grid	Creates row/column layout
position	Controls element position
z-index	Controls layer order

Property	Use
opacity	Controls transparency
transition	Makes style changes smooth
transform	Moves, rotates, or scales element
animation	Creates animation
@media	Makes responsive design

23. Beginner Tips

- Use external CSS for real projects.
- Use class selectors more than ID selectors.
- Keep CSS clean and organized.
- Use proper spacing with margin and padding.
- Make every design responsive for mobile screens.
- Do not use too many colors in one design.
- Practice by designing buttons, cards, forms, navbars, and profile pages.

24. Simple Practice Task: Profile Card

Create a profile card using HTML and CSS. This practice teaches selectors, image styling, card design, shadow, spacing, button styling, and hover effect.

```
<!DOCTYPE html>
<html>
<head>
  <title>Profile Card</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #eef2f5;
    }

    .profile-card {
      width: 300px;
      background-color: white;
      margin: 50px auto;
      padding: 25px;
      text-align: center;
      border-radius: 15px;
      box-shadow: 0 5px 20px rgba(0,0,0,0.1);
    }

    .profile-card img {
      width: 120px;
      height: 120px;
      border-radius: 50%;
    }

    .profile-card button {
      background-color: #007bff;
      color: white;
      padding: 10px 18px;
      border: none;
      border-radius: 8px;
      cursor: pointer;
    }

    .profile-card button:hover {
      background-color: #0056b3;
    }
  </style>
</head>
<body>
  <div class="profile-card">
    
    <h2>Ali Khan</h2>
    <p>Frontend Developer</p>
    <button>Contact Me</button>
  </div>
</body>

</html>
```